

⌘ Virtual Private Networks (VPNs) – In-Depth Theory

⌘ 1. Introduction to VPN

A **VPN (Virtual Private Network)** creates a secure, encrypted tunnel over a public or shared network (like the Internet) to allow private communication between users and resources. It:

- Provides **confidentiality, integrity, and authenticity** of data
 - Allows remote users to securely access **corporate or private networks**
 - Masks IP addresses, **hiding user activity and location**
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⌘ 2. VPN Protocols & Their Characteristics

Various **VPN protocols** define how data is encapsulated, encrypted, and transmitted securely. Each protocol has unique strengths.

Protocol	Characteristics	Use Cases
PPTP (Point-to-Point Tunneling Protocol)	Fast, easy to set up, outdated encryption	Legacy, low-security VPNs
L2TP/IPSec	Layer 2 Tunneling + strong encryption via IPSec	Secure site-to-site VPNs
IPSec	Encrypts at the IP layer, used in site-to-site VPNs	Enterprise WAN links
SSL/TLS VPN	Uses HTTPS, no client required	Secure access via browser
OpenVPN	Open-source, flexible, uses SSL/TLS, highly secure	Widely used remote access
IKEv2/IPSec	Stable, reconnects well, good for mobile	Mobile VPNs
WireGuard	Lightweight, modern cryptography, very fast	Replacing older protocols

🔑 **Key Features of Protocols:**

Feature	PPTP	L2TP/IPSec	OpenVPN	IKEv2	WireGuard
Encryption	Weak	Strong	Strong	Strong	Very strong
Speed	Fast	Moderate	Moderate	Fast	Very fast
Firewall Bypass	Poor	Good	Excellent	Good	Excellent

Stability	Low	Moderate	High	High	High
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☒ 3. VPN Functions

VPNs offer the following **core functionalities**:

- **Confidentiality** – Encrypting data
- **Authentication** – Verifying user identity
- **Data Integrity** – Ensuring data isn't tampered with
- **Tunneling** – Encapsulating data in packets

- **Anonymity** – Hiding real IP address
 - **Remote Access** – Enabling secure connection to private networks from anywhere
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☒ 4. Types of VPN

VPNs can be broadly categorized as:

☒ Remote Access VPN

- For individual users
- Allows connection to private network remotely

- Common in work-from-home setups

☒ *Example:* An employee connects securely to the office network using OpenVPN.

☒ Site-to-Site VPN

- Connects entire networks together (e.g., branch offices)
- Used in corporate WAN setups
- Typically configured between firewalls or routers

☒ *Example:* Two company locations sharing data over IPSec.

☒ Intranet VPN

- Used within a single organization to connect multiple internal locations securely.
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☒ Extranet VPN

- Connects an organization with third-party suppliers or partners
 - Controlled access is given
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☒ 5. Secure VPN

Secure VPN is a broad term for VPNs that offer strong encryption and authentication.

Features include:

- AES-256 encryption
- TLS/SSL for authentication
- Perfect Forward Secrecy (PFS)
- No-log policies
- Leak protection (DNS, WebRTC, IPv6)

☒ Example tools:

- OpenVPN
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WireGuard

- IKEv2/IPSec

Secure VPNs are crucial in environments handling:

- Financial transactions
- Intellectual property
- Medical records (HIPAA compliance)

⌘ 6. Trusted VPN

A Trusted VPN:

- Operates over a **trusted carrier network** or MPLS infrastructure
- Does **not encrypt** traffic by default
- Uses **logical or physical isolation** for security

☒ *Used in:*

- Telecoms, banking, or enterprise backbones
- MPLS-based WANs

☒ **Secure VPN vs Trusted VPN**

Feature	Secure VPN	Trusted VPN
Encryption	Yes (strong)	Often No (relies on isolation)
Transport	Public Internet	MPLS/private network
Use case	Internet-based access	Inter-office secure comms
Security mechanism	Cryptographic tunneling	Trusted routing paths